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IMPACT PARAMETER REPRESENTATION FROM THE WATSON-SOMMERFELD TRANSFORM

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Using the Watson-Sommerfeld transform the elastic scattering amplitude of two spinless particles is shown to have an exact and unique impact parameter, or Fourier-Bessel (FB) representation The representation is valid for all physical energies and scattering angles. Wallace's recent work is found to be an asymptotic expansion of the FB amplitude obtained from the partial-wave expansion The way singularities of the partial-wave amplitude in the l-plane enter in the FB amplitude is also explicitly shown

1 Introduction

It has now become evident from various phenomenological investigations that the impact parameter representation provides an effective tool to explore high-energy particle and nuclear scattering [1–4] However, the representation has not been put on a firm foundation even for the simplest case of elastic scattering by two spinless particles In relativistic theory the impact parameter representation has been obtained by summing infinite sets of Feynman diagrams [5–6], and in non-relativistic theory by using the approximate Schrodinger wave function [7] and by modifying the propagator in the Born series expansion [8] Because all the above derivations are approximate in nature, the impact parameter representation has traditionally been considered as semiclassical, approximate and reliable only for high-energy small-angle scattering

The possibility of writing down an exact impact parameter or a Fourier-Bessel (FB) representation for the elastic scattering amplitude valid for all energies and scattering angles was noted by Cottingham and Peierls nearly a decade ago [9] This was followed by the extensive work of Adachi and Kotani [10] and of Predazzi [11] However, it was pointed out that the FB amplitude derived in this fashion from partial wave expansion was very arbitrary [12] The arbitrariness originates from the fact that the behavior of the scattering amplitude in the whole unphysical momentum transfer region is not specified Thus the question of establishing an ex-

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act, unambiguous FB representation for the elastic scattering amplitude has remained open for the last several years, even though it was recognized early in the development that the impact parameter description was a powerful tool

Recently some interesting results have been reported by Wallace [13] regarding the FB representation of the scattering amplitude. He indicates that the partial wave sum can be converted without approximation to a FB integral The corresponding FB amplitude is connected by a complex derivative operator to the partial wave interpolating function Most important, he obtains systematic corrections in p^{-2} (p = c m momentum) to the usual semiclassical result However, Wallace's derivation is basically a mathematical "tour de force", and does not answer questions such as, (a) why should the result be considered a high-energy expansion as opposed to an exact expansion of the FB amplitude? (b) does his work remove the arbitrariness of the FB amplitude mentioned above? To make his results more transparent, Wallace in a later paper [14] has shown that the complex derivative operator obtained by him provides the standard connection between the FB amplitude and the partial-wave amplitude expected from unsubtracted momentum transfer dispersion relations [15] While this in some sense strengthens his earlier derivation, it does not answer questions of the nature raised above

Next we turn to a dynamical question If the FB amplitude has to be physically meaningful, then it must contain the dynamics of two-body interactions In particular, the energy levels of the interacting system (that is, the bound states and resonances occurring in various partial waves) must enter in the FB amplitude How does this happen? From the theory of complex angular momentum [16–18] we know that the energy levels are represented by Regge poles So what we would like to know is how Regge poles eneter in the construction of the FB amplitude This question has assumed greater significance in view of recent analyses of two-body inelastic high-energy reactions [19] These extensive phenomenological analyses indicate that the geometrical ideas of impact parameter language are intimately connected with Regge poles in the direct channel

In the context of the above remarks we address ourselves to the following three questions in this paper

(i) Can an exact unique FB representation be derived valid for all energies and scattering angles?

(ii) Can one understand better the nature of the complex derivative operator found by Wallace that provides systematic high-energy corrections to the semiclassical result?

(iii) How do the singularities of the partial wave amplitude in the complex angular momentum plane enter in the construction of the FB amplitude?

2. Wallace's expansion of the Fourier-Bessel amplitude

For our purpose it will turn out to be convenient to examine first the second question posed above Our starting point is the Watson-Sommerfeld transform

[16–18] that expresses the scattering amplitude as a contour integral of the analytically continued partial wave amplitude

$$T(s,z) = -\frac{1}{2\pi i}\frac{W}{p}\int_C \mathrm{d}l\, a(l,s)(2l+1)P_l(2y^2-1)\frac{\pi}{\sin\pi l} \tag{2 1}$$

The contour C goes around the real axis clockwise enclosing all the poles at integral values of l ($l = 0,1,2,\ \ldots$) In (2 1) $a(l,s)$ represents the analytically continued partial-wave amplitude, s the square of the c m energy and $z = \cos\theta$ is the c m. scattering angle From now on instead of z we shall use the variable $y = \sin\frac{1}{2}\theta$, so that $z = 1-2y^2$ The physical scattering region corresponds to $0 \leqslant y < 1$ In all our discussions s is taken above threshold

We let the contour C collapse on the real axis and write eq (2 1) as

$$T(s,y) = -\frac{1}{2\pi i}\frac{W}{p}\int_{l_0}^{\infty} \mathrm{d}l\, a(l,s)(2l+1)P_l(2y^2-1)\left\{\frac{\pi}{\sin\pi l_+}-\frac{\pi}{\sin\pi l_-}\right\}, \tag{2 2}$$

where $l_\pm = l \pm i\epsilon$ and $0 > l_0 > -1$ For scattering in the physical region ($y < 1$), the distribution

$$D(l,y) \equiv -(2l+1)P_l(2y^2-1)\left\{\frac{\pi}{\sin\pi l_+}-\frac{\pi}{\sin\pi l_-}\right\} \tag{2 3}$$

can be expressed as a FB transform,

$$D(l,y) = 2\pi i\sum_{n=0}^{\infty}(2n+1)P_n(1-2y^2)\,\delta(l-n)$$

$$= 2\pi i\int_0^{\infty}\beta\mathrm{d}\beta\, J_0(\beta y)\sum_{n=0}^{\infty}(2n+1)\frac{J_{2n+1}(\beta)}{\beta}\,\delta(l-n)\,, \qquad (l > -1) \tag{2 4}$$

To obtain (2 4) we have used the relation

$$\int_0^{\infty}\mathrm{d}\beta\, J_0(\beta y)\, J_{2n+1}(\beta) = P_n(1-2y^2)\theta(1-y)\,, \tag{2 5}$$

where $\theta(x)$ is the step function Inserting eq (2 4) in eq (2 2) we get for $0 \leqslant y < 1$,

$$T(s,y) = \frac{W}{2p}\int_0^{\infty}\beta\mathrm{d}\beta\, J_0(\beta y)\, h_1(s,\beta)\,, \tag{2 6}$$

where

$$h_1(s,\beta) = 2 \int_{l_0}^{\infty} \mathrm{d}l\, a(l,s) \sum_{n=0}^{\infty} (2n+1) \frac{J_{2n+1}(\beta)}{\beta} \delta(l-n) \tag{2 7}$$

$$= 2 \sum_{n=0}^{\infty} (2n+1) \frac{J_{2n+1}(\beta)}{\beta} a_n(s) \tag{2 8}$$

Eqs (2 6) and (2 8) correspond to the usual impact parameter representation obtained from partial-wave expansion [9, 10] The variable β has to be identified as $\beta = 2pb$, where p is the c m momentum and b is the impact parameter [12] As seen from (2 8), the FB amplitude in this case is an infinite discrete sum over the partial-wave amplitudes

We now derive Wallace's result First we point out that the infinite sum of δ-functions in eq (2 7) can be expressed as

$$\sum_{n=0}^{\infty} (2n+1) \frac{J_{2n+1}(\beta)}{\beta} \delta(l-n) = \frac{1}{2\pi i} [\mathcal{F}(l,\beta) + \mathcal{F}(-l-1,\beta)], \qquad (l>-1), \tag{2 9}$$

where

$$\mathcal{F}(l,\beta) = \frac{l+\frac{1}{2}}{\beta} \int_{\gamma-i\infty}^{\gamma+i\infty} \mathrm{d}\zeta\, \mathrm{e}^{-\zeta(l+1/2)+\beta \sinh \zeta/2} \tag{2 10}$$

The proof is given in appendix A Using (2 9) in (2 7) we find that $h_1(s,\beta)$ can be written as

$$h_1(s,\beta) = \frac{1}{\pi i} \int_{l_0}^{\infty} \mathrm{d}l\, a(l,s)\, [\mathcal{F}(l,\beta) + \mathcal{F}(-l-1,\beta)] \tag{2 11}$$

Next we show that $\exp(\beta \sinh \frac{1}{2}\zeta)$ can be expressed as a derivative operator operating on $\exp(\beta \frac{1}{2}\zeta)$,

$$\mathrm{e}^{\beta \sinh \zeta/2} = \sum_{n=0}^{\infty} \frac{\beta^n}{n!} (\sinh \tfrac{1}{2}\zeta)^n$$

$$= \sum_{n=0}^{\infty} \frac{\beta^n}{n!} (\tfrac{1}{2}\zeta)^n \left(\frac{\frac{1}{2}\zeta}{\sinh \frac{1}{2}\zeta}\right)^{-n} \tag{2 12a}$$

$$= \sum_{n=0}^{\infty} \frac{\beta^n}{n!} (\tfrac{1}{2}\zeta)^n \sum_{k=0}^{\infty} b_k(-\tfrac{1}{2}n) \frac{\zeta^{2k}}{(2k)!} \tag{2 12b}$$

$$= \sum_{k=0}^{\infty} \frac{\zeta^{2k}}{(2k)!} \sum_{n=0}^{\infty} \frac{1}{n!} b_k(-\tfrac{1}{2}n) \beta^n (\tfrac{1}{2}\zeta)^n$$

$$= \sum_{k=0}^{\infty} \frac{\zeta^{2k}}{(2k)!} \sum_{n=0}^{\infty} \frac{1}{n!} b_k\left(-\tfrac{1}{2}\beta\frac{\mathrm{d}}{\mathrm{d}\beta}\right) \beta^n (\tfrac{1}{2}\zeta)^n$$

$$= \sum_{k=0}^{\infty} \frac{\zeta^{2k}}{(2k)!} b_k\left(-\tfrac{1}{2}\beta\frac{\mathrm{d}}{\mathrm{d}\beta}\right) \mathrm{e}^{\beta\zeta/2}$$

$$= \sum_{k=0}^{\infty} \frac{1}{(2k)!} b_k\left(-\tfrac{1}{2}\beta\frac{\mathrm{d}}{\mathrm{d}\beta}\right)\left(2\frac{\mathrm{d}}{\mathrm{d}\beta}\right)^{2k} \mathrm{e}^{\beta\zeta/2} \tag{2 12c}$$

To go from (2 12a) to (2 12b) we have used the following expansion

$$\left(\frac{\frac{1}{2}\zeta}{\sinh\frac{1}{2}\zeta}\right)^{2x} = \sum_{k=0}^{\infty} B_{2k}^{(2x)}(x)\frac{\zeta^{2k}}{(2k)!} \qquad (|\zeta| < 2\pi)\,,$$

where $B_{2k}^{(2x)}(x)$ are generalized Bernoulli polynomials [20] ($b_k(x) \equiv B_{2k}^{(2x)}(x)$ in Wallace's notation) In eq (2 12b) the restriction $|\zeta| < 2\pi$ does not occur, because $((\sinh\frac{1}{2}\zeta)/\frac{1}{2}\zeta)^n$ is an entire function of ζ when n is a positive integer Using eq (2 12c) we can then write $\mathcal{F}(l,\beta)$ given by (2 10) as

$$\mathcal{F}(l,\beta) = \frac{l+\frac{1}{2}}{\beta}\int_{\gamma-i\infty}^{\gamma+i\infty} \mathrm{d}\zeta\, \mathrm{e}^{-\zeta(l+1/2)}\,\hat{O}(\beta)\,\mathrm{e}^{\beta\zeta/2}\ , \tag{2 13}$$

where

$$\hat{O}(\beta) \equiv \sum_{k=0}^{\infty} \frac{1}{(2k)!} b_k\left(-\tfrac{1}{2}\beta\frac{\mathrm{d}}{\mathrm{d}\beta}\right)\left(2\frac{\mathrm{d}}{\mathrm{d}\beta}\right)^{2k} \tag{2 14}$$

Let us now define a new distribution $\widetilde{\mathcal{F}}(l,\beta)$ obtained from $\mathcal{F}(l,\beta)$ by formally interchanging the operator $\hat{O}(\beta)$ and the integration in eq (2 13),

$$\widetilde{\mathcal{F}}(l,\beta) \equiv \frac{l+\frac{1}{2}}{\beta}\,\hat{O}(\beta)\int_{\gamma-i\infty}^{\gamma+i\infty} \mathrm{d}\zeta\, \mathrm{e}^{\zeta(\beta/2-l-1/2)} \tag{2 15}$$

Replacing $\mathcal{F}(l,\beta)$ by $\widetilde{\mathcal{F}}(l,\beta)$ in eq (2 11) leads to a new FB amplitude $\tilde{h}_1(s,\beta)$,

$$\tilde{h}_1(s,\beta) \equiv \frac{1}{\pi i}\int_{l_0}^{\infty} \mathrm{d}l\, a(l,s)\, [\widetilde{\mathcal{F}}(l,\beta) + \widetilde{\mathcal{F}}(-l-1,\beta)] \tag{2 16}$$

This amplitude can be easily evaluated Using the following representation of the

δ-function

$$\delta(x) = \frac{1}{2\pi i} \int_{\gamma - i\infty}^{\gamma + i\infty} \mathrm{d}\zeta \, \mathrm{e}^{\zeta x} ,$$

we have from (2 15)

$$\widetilde{\mathcal{F}}(l, \beta) = 2\pi i \frac{1}{\beta} \hat{O}(\beta) \beta \delta(\beta - 2l - 1) \tag{2 17}$$

Inserting (2 17) in (2 16) and choosing $l_0 = -\frac{1}{2}$, we find

$$\widetilde{h}_1(s, \beta) = O\left(\beta, \frac{\mathrm{d}}{\mathrm{d}\beta}\right) a(\tfrac{1}{2}\beta - \tfrac{1}{2}, s) , \tag{2 18}$$

where

$$O\left(\beta, \frac{\mathrm{d}}{\mathrm{d}\beta}\right) = \frac{1}{\beta} \hat{O}(\beta) \beta$$

$$= \frac{1}{\beta} \sum_{k=0}^{\infty} \frac{1}{(2k)!} b_k \left(-\tfrac{1}{2}\beta \frac{\mathrm{d}}{\mathrm{d}\beta}\right) \left(2 \frac{\mathrm{d}}{\mathrm{d}\beta}\right)^{2k} \beta \tag{2 19}$$

The FB amplitude $\widetilde{h}_1(s, \beta)$ related to the partial-wave amplitude $a(l, s)$ *via* eqs (2 18) and (2 19) is precisely the one obtained by Wallace [13] We see that the complex derivative operator operating on $a(l, s)$ originates from the power series expansion (2 12c) Furthermore, if β is put equal to $2pb$, then (2 18) does indeed provide an expansion in inverse powers of p^2 The first term of the expansion gives $\widetilde{h}_1(s, 2pb) \simeq a(pb - \frac{1}{2}, s)$ as we expect semi-classically at high energy

Our arguments above show that if a formal interchange of $\hat{O}(\beta)$ and integration is done in (2 13), then $h_1(s, \beta)$ of eq (2 8) leads to $\widetilde{h}_1(s, \beta)$ of Wallace The question is then Is this formal interchange actually allowed? The answer is no, that is, $\mathcal{F}(l, \beta) \neq \widetilde{\mathcal{F}}(l, \beta)$ This can be demonstrated by a simple example

Let us take $a(l, s) = \mathrm{e}^{-\alpha(l + 1/2)}$ with α real and positive Then

$$h_1(s, \beta) = 2 \sum_{n=0}^{\infty} (2n+1) \frac{J_{2n+1}(\beta)}{\beta} \mathrm{e}^{-\alpha(n+1/2)}$$

$$= 2 \int_0^1 y \mathrm{d}y \, J_0(\beta y) \sum_{n=0}^{\infty} (2n+1) P_n(1 - 2y^2) \, \mathrm{e}^{-\alpha(n+1/2)}$$

$$= \int_0^1 y \mathrm{d}y \, J_0(\beta y) \frac{\sinh \frac{1}{2}\alpha \cosh \frac{1}{2}\alpha}{[\sinh^2 \frac{1}{2}\alpha + y^2]^{3/2}} . \tag{2 20}$$

On the other hand,

$$\widetilde{h}_1(s,\beta) = O\left(\beta, \frac{\mathrm{d}}{\mathrm{d}\beta}\right) \mathrm{e}^{-\alpha\beta/2}$$

$$= \cosh \tfrac{1}{2}\alpha \, \mathrm{e}^{-\beta \sinh(\alpha/2)}$$

$$= \int_0^\infty y\,\mathrm{d}y\, J_0(\beta y) \frac{\sinh \frac{1}{2}\alpha \cosh \frac{1}{2}\alpha}{[\sinh^2 \frac{1}{2}\alpha + y^2]^{3/2}} \tag{2 21}$$

We see that $h_1(s, \beta)$ and $\widetilde{h}_1(s, \beta)$ are different, since one is an integral from 0 to 1 while the other is from 0 to ∞ (It can also be shown that the integral from 1 to ∞ does not vanish) Hence the interchange of $\hat{O}(\beta)$ and integration in (2 13) is not allowed

We may now wonder what relation $\widetilde{h}_1(s, \beta)$ has with $h_1(s, \beta)$ knowing that it is the latter that gives the FB representation of the scattering amplitude in the physical region (eq (2 6)) The clue comes from the above example If β is large, then the dominant contributions to the integrals in eqs (2 20) and (2 21) arise from the small y region because of the oscillations of the Bessel function for large argument Hence $h_1(s, \beta) \to \widetilde{h}_1(s, \beta)$ as $\beta \to \infty$ So we expect $\widetilde{h}_1(s, \beta)$ to be the asymptotic form of $h_1(s, \beta)$

To examine this point more carefully we show in appendix A that

$$\mathcal{F}(l, \beta) = \frac{1}{\beta}\hat{O}(\beta)\; 2i(2l+1) \sum_{n=-\infty}^{\infty} \delta(2l+1-n) \frac{\sin \pi(\beta - 2l - 1)}{\beta - 2l - 1}\ , \tag{2 22}$$

and correspondingly we get from (2 11)

$$h_1(s, \beta) = \frac{1}{\beta}\, \hat{O}(\beta)\, \beta \sum_{n=0}^{\infty} a(\tfrac{1}{2}n - \tfrac{1}{2}, s) \left[\frac{\sin \pi(n-\beta)}{\pi(n-\beta)} + \frac{\sin \pi(n+\beta)}{\pi(n+\beta)}\right] \tag{2 23}$$

Only for $\beta \to \infty$ we find that the infinite sum on the right-hand side approaches $a(\frac{1}{2}\beta - \frac{1}{2}, s)$ This of course means that $\widetilde{h}_1(s, \beta)$ of Wallace (eq (2 18)) is the asymptotic form of $h_1(s, \beta)$, and Wallace's expansion is an asymptotic expansion in inverse powers of p^2 when we put $\beta = 2pb$ and take limit $p \to \infty$

At this point a few clarifying remarks are necessary It may be noticed that for the example $a(l, s) = \mathrm{e}^{-\alpha(l+1/2)}$ chosen above, Wallace's $\widetilde{h}_1(s, \beta)$ given by eq (2 21) is in fact the full FB amplitude One may, therefore, wonder what is the relevance of showing $\widetilde{h}_1(s, \beta)$ to be the asymptotic form of $h_1(s, \beta)$, since the latter is only a part of the full FB amplitude

To answer this point, let us consider the example $a(l, s) = 1/(\alpha - l)(\alpha + l + 1)$ In this case the scattering amplitude is $\pi P_\alpha(2y^2 - 1)/\sin \pi\alpha$, and as we argue in sect 3, the FB amplitude has to be highly singular at the origin ($\sim\beta^{-2\alpha-2}$), and in fact has to be treated as a distribution there This is a reflection of the fact that the amplitude $\pi P_\alpha(2y^2-1)/\sin\pi\alpha$ grows like a power ($\sim y^{2\alpha}$) when $y\to\infty$ On the other hand, Wallace's result for this case,

$$\tilde{h}_1(s,\beta)=\frac{1}{\beta}\sum_{k=0}^{\infty}\frac{1}{(2k)!}\,b_k\left(-\tfrac{1}{2}\beta\frac{\mathrm{d}}{\mathrm{d}\beta}\right)\left(2\frac{\mathrm{d}}{\mathrm{d}\beta}\right)^{2k}\beta\,\frac{4}{(2\alpha+1)^2-\beta^2}\,,$$

gives a FB amplitude which cannot be more singular than $1/\beta$ at $\beta=0$ This illustrates that Wallace's result cannot reproduce the correct FB amplitude, when the scattering amplitude grows like a power and the FB amplitude is highly singular at the origin Of course, even in such a case the usefulness of Wallace's $\tilde{h}_1(s,\beta)$ is that it still provides the asymptotic expansion of the piece $h_1(s,\beta)$ of the full FB amplitude (Incidentally, the power behavior is quite general, since it corresponds to the presence of direct-channel Regge poles, i e , direct-channel bound states and resonances)

3 Derivation of a unique Fourier-Bessel representation

Our discussion of FB representation so far has been restricted to $y<1$, that is to the physical region In this region, we have found that the scattering amplitude can be written as a FB integral, eq (2 6), with the FB amplitude $h_1(s,\beta)$ given by eq (2 8) However, this FB representation is ambiguous, because the FB amplitude is not unique To see this, let us add to $h_1(s,\beta)$ another term $h_2(s,\beta)$ of the form

$$h_2(s,\beta)=\int_1^{\infty} y'\mathrm{d}y'\,J_0(\beta y')\,\lambda(s,y')\,,$$

where $\lambda(s,y')$ is arbitrary [12] The new FB representation

$$T(s,y)=\frac{W}{2p}\int_0^{\infty}\beta\mathrm{d}\beta\,J_0(\beta y)\,h(s,\beta) \tag{3 1}$$

with

$$h(s,\beta)=h_1(s,\beta)+h_2(s,\beta) \tag{3 2}$$

will still reproduce the same scattering amplitude in the physical region, since by construction for $y<1$

$$\int_0^{\infty}\beta\mathrm{d}\beta\,J_0(\beta y)\,h_2(s,\beta)=\int_1^{\infty}\mathrm{d}y'\,\delta(y-y')\,\lambda(s,y')=0 \tag{3 3}$$

The origin of the above non-uniqueness lies in not specifying the scattering amplitude $T(s, y)$ as a function of y in the unphysical region $1 < y < \infty$ If $T(s, y)$ is known in this region, then $h_2(s, \beta)$ will be fixed by the requirement that its FB integral should reproduce $T(s, y)$ for $\infty > y > 1$ and of course be zero for $y < 1$ (Incidentally, the FB transform of $h_1(s, \beta)$, eq (2 8), always vanishes for $y > 1$, because of eq (2 5)) The new element in our approach is to use the Watson-Sommerfeld transform to specify $T(s, y)$ for $\infty > y > 1$, thus determine $h_2(s, \beta)$ and remove the arbitrariness in the FB representation

There is, however, a mathematical difficulty involved in carrying out this program To see this, let us note that we want to write the scattering amplitude in the form

$$T(s, y) = \frac{W}{2p} \int_0^\infty \beta \mathrm{d}\beta \, J_0(\beta y) \, h_2(s, \beta) \tag{3 4}$$

for $y > 1$ But for s fixed in the physical region, $T(s, y)$ for large y can behave like a power For example, presence of a direct channel Regge pole will imply $T(s, y)$ behaving as $y^{2\alpha(s)}$ for large y To obtain such asymptotic behavior in y, $h_2(s, \beta)$ in (3 4) has to be highly singular at $\beta = 0$ However, if $h_2(s, \beta)$ is too singular, then the FB integral (3 4) does not exist The answer to this problem is that $h_2(s, \beta)$ is to be considered a distribution such that when the FB integral is taken, it reproduces the $T(s, y)$ given by the Watson-Sommerfeld transform

To construct $h_2(s, \beta)$, we first notice that for $-1 < \mathrm{Re}\, l < 0$ and $y > 1$, $P_l(2y^2 - 1)$ has a FB transform [22]

$$\begin{aligned} P_l(2y^2 - 1) &= \frac{1}{2 \cos \pi l} \left[\frac{2}{\pi} \sin \pi l \, Q_l(2y^2 - 1) - \frac{2}{\pi} \sin \pi l \, Q_{-l-1}(2y^2 - 1) \right] \\ &= \int_0^\infty \beta \mathrm{d}\beta \, J_0(\beta y) \frac{1}{2 \cos \pi l} \left[-\frac{J_{2l+1}(\beta)}{\beta} + \frac{J_{-2l-1}(\beta)}{\beta} \right] \end{aligned} \tag{3 5}$$

This result, however, cannot be extended to $\mathrm{Re}\, l > 0$, because the singular behavior of $J_{-2l-1}(\beta)$ at the origin ($J_{-2l-1}(\beta) \sim \beta^{-2l-1}$ as $\beta \to 0$) prevents the existence of the integral We therefore need a way of smoothing this singular behavior, so that the FB integral is defined and reproduces $P_l(2y^2 - 1)$, which behaves as y^{2l} for y large Our result is that $J_{-2l-1}(\beta)/\beta$ in (3 5) is to be replaced by a distribution

$$\Delta(l, \beta^2) \equiv \lim_{\delta \to 0} \beta^{2l+1} J_{-2l-1}(\beta) \left[\frac{\Gamma(-l)}{2\pi i} \int_\infty^{0^+} \frac{\mathrm{d}\lambda}{\lambda^2} (-\lambda)^{-l} \, \mathrm{e}^{-\beta^2/(\lambda+\delta)} \right] \tag{3 6a}$$

$$= \lim_{\delta \to 0} \beta^{2l+1} J_{-2l-1}(\beta) \left[\frac{\Gamma(-l)}{\Gamma(l+2)} \frac{1}{\beta^{2(l+1)}} \int_0^\infty \mathrm{d}x \, J_1(x) \, \mathrm{e}^{-\delta x^2/4\beta^2} (\tfrac{1}{4} x^2)^{l+1} \right] \tag{3 6b}$$

The limit $\delta \to 0$ is of course to be taken after the FB integration is carried out We note that $\beta^{2l+1} J_{-2l-1}(\beta)$ is a well-behaved power series in β^2 being finite at the origin The singularity β^{-2l-1} expected from eq (3 5) at the origin is contained in the square brackets Eq (3.6a) shows how the singularity is handled at the origin, while eq (3 6b) exhibits explicitly the smoothing done at $\beta^2 = 0$ For Re $l > 0$ and $y > 1$, we find

$$P_l(2y^2 - 1) = \int_0^\infty \beta \mathrm{d}\beta \, J_0(\beta y) \frac{1}{2\cos \pi l} \left[\Delta(l, \beta^2) - \frac{J_{2l+1}(\beta)}{\beta}\right] \tag{3 7}$$

Detailed proof of this is given in appendix B

Use of eq (3 7) in the Watson-Sommerfeld transform gives us for $y > 1$,

$$T(s, y) = \frac{W}{2p} \int_0^\infty \beta \mathrm{d}\beta \, J_0(\beta y) \, h_2(s, \beta) , \tag{3 8}$$

where

$$h_2(s, \beta) = -\frac{1}{2\pi i} \int_C \mathrm{d}l \, a(l, s)(2l + 1) \frac{\pi}{\sin \pi l} \frac{1}{\cos \pi l} \left[\Delta(l, \beta^2) - \frac{J_{2l+1}(\beta)}{\beta}\right] \tag{3 9}$$

In appendix B, we further show that for $y < 1$,

$$\int_0^\infty \beta \mathrm{d}\beta \, J_0(\beta y) \, \Delta(l, \beta^2) = P_l(1 - 2y^2) \tag{3 10}$$

Combining this with the result

$$\int_0^\infty \beta \mathrm{d}\beta \, J_0(\beta y) \frac{J_{2l+1}(\beta)}{\beta} = P_l(1 - 2y^2) \quad (y < 1) , \tag{3 11}$$

we find for $y < 1$

$$\int_0^\infty \beta \mathrm{d}\beta \, J_0(\beta y) \, h_2(s, \beta) = 0 . \tag{3 12}$$

Our final result is then

$$T(s, y) = \frac{W}{2p} \int_0^\infty \beta \mathrm{d}\beta \, J_0(\beta y) \, h(s, \beta) \tag{3 13}$$

for $\infty > y \geqslant 0$ and

$$h(s, \beta) = h_1(s, \beta) + h_2(s, \beta) , \tag{3 14}$$

where $h_1(s, \beta)$ is given by eq (2 8) and $h_2(s, \beta)$ by eq (3 9) The part $h_1(s, \beta)$ of the FB amplitude determines the amplitude $T(s, y)$ for y in the physical region ($0 \leqslant y < 1$) The part $h_2(s, \beta)$ determines $T(s, y)$ for $\infty > y > 1$ It is the determination of the latter part that has removed the arbitrariness in defining the FB amplitude.

That $h(s, \beta)$ in eq (3 13) is unique can be explicitly shown Suppose we have another FB amplitude $h'(s, \beta)$ that reproduces $T(s, y)$ for all y ($\infty > y \geqslant 0$) Then we must have

$$\int_0^\infty \beta \mathrm{d}\beta \, J_0(\beta y) \, [h(s, \beta) - h'(s, \beta)] = 0$$

for all y But this equation means

$$h(s, \beta) - h'(s, \beta) = 0$$

Hence $h(s, \beta)$ is unique

4. Fourier-Bessel amplitude and singularities in the l-plane

We now investigate how the l-plane analytic properties of the partial-wave amplitude $a(l, s)$ enter in the FB amplitude Let us consider first $h_1(s, \beta)$ which determines the scattering amplitude in the physical region Using the formula

$$\int_0^1 y \mathrm{d}y \, J_0(\beta y) \, P_n(1 - 2y^2) = \frac{J_{2n+1}(\beta)}{\beta} \tag{4 1}$$

in eq (2 7) we get

$$\begin{aligned} h_1(s, \beta) &= 2 \int_0^1 y \mathrm{d}y \, J_0(\beta y) \int_{l_0}^\infty \mathrm{d}l \, a(l, s) \sum_{n=0}^\infty (2n + 1) \, P_n(1 - 2y^2) \delta(l - n) \\ &= -2 \int_0^1 y \mathrm{d}y \, J_0(\beta y) \frac{1}{2\pi i} \int_C \mathrm{d}l \, a(l, s)(2l + 1) \frac{\pi P_l(2y^2 - 1)}{\sin \pi l} . \end{aligned} \tag{4 2}$$

For $0 \leqslant y < 1$, $P_l(2y^2 - 1)$ has a partial-wave expansion

$$P_l(2y^2 - 1) = \sum_{n=0}^\infty (2n + 1) P_n(1 - 2y^2) \frac{\sin \pi l}{\pi(l - n)(l + n + 1)} \tag{4 3}$$

Inserting this in (4 2) and using (4 1) again we obtain

$$h_1(s,\beta) = -2\sum_{n=0}^{\infty}(2n+1)\frac{J_{2n+1}(\beta)}{\beta}\frac{1}{2\pi i}\int_C dl\,\frac{a(l,s)(2l+1)}{(l-n)(l+n+1)} \tag{4 4}$$

Eq (4 4) shows how the singularities of the partial wave amplitude show up in $h_1(s,\beta)$ Displacing the contour C away from the real axis exposes the poles and cuts of $a(l,s)$, and the contribution from any singularity can be explicitly written down For example, a Regge-pole term $r(s)/(l-\alpha(s))$ in $a(l,s)$ gives the following contribution to $h_1(s,\beta)$

$$h_1^{\mathrm{R}}(s,\beta) = -2\sum_{n=0}^{\infty}(2n+1)\frac{J_{2n+1}(\beta)}{\beta}\frac{r(s)(2\alpha+1)}{(\alpha-n)(\alpha+n+1)} \tag{4 5}$$

The FB transform of $h_1^{\mathrm{R}}(s,\beta)$ is indeed the usual Regge pole amplitude for $y<1$,

$$\frac{W}{2p}\int_0^{\infty}\beta d\beta\, J_0(\beta y)\, h_1^{\mathrm{R}}(s,\beta)$$

$$= -\frac{W}{p}\sum_{n=0}^{\infty}(2n+1)P_n(1-2y^2)\frac{r(s)(2\alpha+1)}{(\alpha-n)(\alpha+n+1)}\,\theta(1-y)$$

$$= -\frac{W}{p}r(s)(2\alpha+1)\frac{\pi}{\sin\pi\alpha}P_\alpha(2y^2-1)\theta(1-y) \qquad \text{(using eq (4 3))} \tag{4 6}$$

The FB amplitude $h_2(s,\beta)$ that determines the scattering amplitude in the unphysical region $(1<y<\infty)$ has already been expressed as a contour integral of $a(l,s)$ in the l-plane (eq (3 9)) So the contributions from the singularities of $a(l,s)$ to $h_2(s,\beta)$ are obtained, as in the case of $h_1(s,\beta)$, by displacing the contour C and exposing the singularities In particular, the single Regge pole term $r(s)/(l-\alpha(s))$ in $a(l,s)$ gives the contribution

$$h_2^{\mathrm{R}}(s,\beta) = -r(s)\frac{(2\alpha+1)\pi}{\sin\pi\alpha}\frac{1}{\cos\pi\alpha}\left[\Delta(\alpha,\beta^2)-\frac{J_{2\alpha+1}(\beta)}{\beta}\right] \tag{4 7}$$

The FB amplitude of this term is as expected the single Regge pole amplitude for $y>1$

$$\frac{W}{2p}\int_0^{\infty}\beta d\beta\, J_0(\beta y)\, h_2^{\mathrm{R}}(s,\beta) = -\frac{W}{p}r(s)\frac{(2\alpha+1)\pi}{\sin\pi\alpha}P_\alpha(2y^2-1)\theta(y-1)\,,$$

$$\text{(using eq (B 5))} \tag{4 8}$$

5 Concluding remarks

We first conclude from eq (3 13) that an exact unique FB or impact-parameter representation of the scattering amplitude exists valid for all energies and scattering angles ($0 \leqslant \theta \leqslant \pi$) The non-uniqueness of the FB amplitude in previous formulations [12] arose from not specifying the behavior of the scattering amplitude in the unphysical momentum transfer region ($-\infty < t < -4p^2$) We have circumvented this difficulty by specifying the scattering amplitude for all momentum transfers by the Watson-Sommerfeld transform Of course the price we have paid is that the FB amplitude is now a distribution at the origin*

We have shown that what Wallace has obtained is an asymptotic expansion of $h_1(s, \beta)$, the part of FB amplitude that determines the physical scattering ($0 \leqslant \theta < \pi$) Of course his derivation and arguments do not indicate this The merit of Wallace's work lies in the fact that this expansion of $h_1(s, \beta)$ with the semi-classical result as the leading term is well suited for computation, on the other hand, it is not realistic to try to compute the exact $h_1(s, \beta)$ given by eq (2 8) because of the rapid oscillations of the Bessel functions

By using the Watson-Sommerfeld transform, we have been able to express $h(s, \beta)$ as a contour integral over $a(l, s)$ in the l-plane This not only expresses in a formal way the close relationship between the FB amplitude and the partial-wave amplitude, but also allows to incorporate easily contributions from singularities of $a(l, s)$ into $h(s, \beta)$ In particular, if we know of a set of bound states and resonances lying on a Regge trajectory in the s-channel, then we can explicitly construct the corresponding piece of the FB amplitude**.

The basic assumption of our investigation is the analyticity of the partial-wave amplitude around the positive real axis in the l-plane This assumption appears now to be on solid foundation in non-relativistic [16, 17] as well as in relativistic scattering [18] We can, therefore, use the impact parameter representation as an exact tool no longer limited by any high-energy small-angle approximation

* That the FB amplitude has to be a distribution if the scattering amplitude behaves like a power has been noted by Kupsch and Stamatescu [26] These authors study the analytic properties of the FB amplitude in the impact parameter plane given that the scattering amplitude satisfies a subtracted dispersion relation in the momentum transfer variable t They also examine FB amplitudes obtained from scattering amplitudes with cut-off in t [27]
Perhaps it is useful to note that while $h_2(s, \beta)$ in general should be a distribution, in cases where the scattering amplitude $T(s, y)$ is suitably bounded ($\int_0^\infty y^{1/2} |T(s, y)| \mathrm{d}y < \infty$) it becomes an ordinary function This corresponds to the existence of the usual FB representation via Hankel's theorem [12]

** To keep the present treatment simple, we have not included the signature property of the analytically continued partial-wave amplitude This of course can be easily done by writing Watson-Sommerfeld transforms for the even and the odd partial wave series and then proceeding the way we have

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Appendix A

In this appendix, we give the proof of eq (2.9),

$$
\begin{aligned}
&2\pi i \sum_{n=0}^{\infty} (2n+1)\frac{J_{2n+1}(\beta)}{\beta}\,\delta(n-l) \\
&= 4\pi i \sum_{n=0}^{\infty} (2n+1)\frac{J_{2n+1}(\beta)}{\beta}\,\delta(2n+1-2l-1) \\
&= 4\pi i \frac{(2l+1)}{\beta} \sum_{m=1,3,5,}^{\infty} J_m(\beta)\,\delta(m-2l-1) \\
&= 2\pi i \frac{(2l+1)}{\beta} \sum_{m=0}^{\infty} [J_m(\beta)-J_{-m}(\beta)]\,\delta(m-2l-1) \\
&= 2\pi i \frac{(2l+1)}{\beta} \sum_{m=-\infty}^{\infty} [J_m(\beta)-J_{-m}(\beta)]\,\delta(m-2l-1) \qquad (l>-1, 2l+1>-1) \\
&= 2\pi i \frac{(2l+1)}{\beta} \Big[\sum_{m=-\infty}^{\infty} J_m(\beta)\,\delta(m-2l-1) - \sum_{m=-\infty}^{\infty} J_m(\beta)\,\delta(m+2l+1)\Big] \\
&= \frac{(2l+1)}{\beta}\Big[\sum_{m=-\infty}^{\infty} J_m(\beta) \int_{\gamma-i\infty}^{\gamma+i\infty} d\zeta\, e^{(m-2l-1)\zeta} - \sum_{m=-\infty}^{\infty} J_m(\beta) \int_{\gamma-i\infty}^{\gamma+i\infty} d\zeta\, e^{(m+2l+1)\zeta}\Big] \\
&= \frac{(2l+1)}{\beta}\Big[\int_{\gamma-i\infty}^{\gamma+i\infty} d\zeta\, e^{-(2l+1)\zeta} \sum_{m=-\infty}^{\infty} J_m(\beta) e^{m\zeta} \\
&\qquad - \int_{\gamma-i\infty}^{\gamma+i\infty} d\zeta\, e^{(2l+1)\zeta} \sum_{m=-\infty}^{\infty} J_m(\beta)\, e^{m\zeta}\Big] \qquad \text{(A 1)}
\end{aligned}
$$

In eq (A 1) the summations and the integrations have been interchanged The interchanges are justified, because the infinite sum $\sum_{m=-\infty}^{\infty} J_m(\beta)\, e^{m\zeta}$ converges uniformly for $\mathrm{Re}\,\zeta$ finite This can be easily seen by noticing that

$$\sum_{m=-\infty}^{\infty} J_m(\beta)\, t^m = \exp[\tfrac{1}{2}\beta(t-1/t)]$$

converges everywhere except at $|t| = 0$ and $|t| = \infty$ From (A 1) we now obtain

$$2\pi i \sum_{n=0}^{\infty} (2n+1) \frac{J_{2n+1}(\beta)}{\beta} \delta(n-l) \qquad (l > -1)$$

$$= \frac{(2l+1)}{\beta} \left[\int_{\gamma - i\infty}^{\gamma + i\infty} d\zeta \exp(-(2l+1)\zeta + \beta \sinh \zeta) \right.$$

$$\left. - \int_{\gamma - i\infty}^{\gamma + i\infty} d\zeta \exp((2l+1)\zeta + \beta \sinh \zeta) \right]$$

$$= \mathcal{F}(l, \beta) + \mathcal{F}(-l-1, \beta), \tag{A 2}$$

if the variable ζ is changed to $\frac{1}{2}\zeta$ The term $\mathcal{F}(l, \beta)$ is defined by eq (2 10) in the text

To derive eq (2 22) we write $\mathcal{F}(l, \beta)$ in the form

$$\mathcal{F}(l, \beta) = \frac{2l+1}{\beta} \left[\int_{\gamma - i\infty}^{\gamma - i0} d\zeta \exp(-\zeta(2l+1+i\epsilon) + \beta \sinh \zeta) \right.$$

$$\left. + \int_{\gamma + i0}^{\gamma + i\infty} d\zeta \exp(-\zeta(2l+1-i\epsilon) + \beta \sinh \zeta) \right] \tag{A 3}$$

That eq (A.3) is correct can be checked by using the expansion $\exp[\beta \sinh \zeta] = \sum_{n=-\infty}^{\infty} J_n(\beta)\, e^{n\zeta}$ and carrying out the integrations in (A 3) This gives

$$\mathcal{F}(l, \beta) = \frac{(2l+1)}{\beta} \sum_{n=-\infty}^{\infty} J_n(\beta) \left[\frac{\exp(\gamma(n-2l-1))}{n-2l-1-i\epsilon} - \frac{\exp(\gamma(n-2l-1))}{n-2l-1+i\epsilon} \right]$$

$$= 2\pi i \frac{(2l+1)}{\beta} \sum_{n=-\infty}^{\infty} J_n(\beta) \delta(n-2l-1), \tag{A 4}$$

which is indeed the first term on the right-hand side of (A 1) Choosing $\gamma = 0$ and dividing the integration region into intervals of $i2\pi$ we rewrite eq (A 3) as

$$\mathcal{F}(l, \beta) = \frac{2l+1}{\beta} \left[\sum_{n=1}^{\infty} \int_{i2n\pi - i\pi}^{i2n\pi + i\pi} d\zeta \exp(-\zeta(2l+1-i\epsilon) + \beta \sinh \zeta) \right.$$

$$+ \sum_{n=1}^{\infty} \int_{-i2n\pi - i\pi}^{-i2n\pi + i\pi} d\zeta \exp(-\zeta(2l+1+i\epsilon) + \beta \sinh \zeta)$$

$$+\int_{-i\pi}^{i\pi} d\zeta \exp(-\zeta(2l+1)+\beta \sinh \zeta)]$$

$$=\frac{2l+1}{\beta}[\sum_{n=1}^{\infty} \exp(-i2n\pi(2l+1-i\epsilon))+\sum_{n=1}^{\infty} \exp(i2n\pi(2l+1+i\epsilon))+1]$$

$$\times \int_{-i\pi}^{i\pi} d\zeta \exp(-\zeta(2l+1)+\beta \sinh \zeta)$$

$$=\frac{2l+1}{\beta}\left[\frac{1}{1-\exp(-i2\pi(2l+1-i\epsilon))}+\frac{\exp(i2\pi(2l+1+i\epsilon))}{1-\exp(i2\pi(2l+1+i\epsilon))}\right]$$

$$\times \hat{O}(\beta) \int_{-i\pi}^{i\pi} d\zeta \exp(\zeta(\beta-2l-1)) \qquad (\text{using eq (2 12c)})$$

$$=\frac{2l+1}{\beta} \exp(i\pi(2l+1))\left[\frac{1}{\sin \pi(2l+1-i\epsilon)}-\frac{1}{\sin \pi(2l+1+i\epsilon)}\right]$$

$$\times \hat{O}(\beta) \frac{\sin \pi(\beta-2l-1)}{\beta-2l-1}$$

$$=2i \frac{(2l+1)}{\beta} \sum_{n=-\infty}^{\infty} \delta(2l+1-n) \hat{O}(\beta) \frac{\sin \pi(\beta-2l-1)}{\beta-2l-1} \tag{A 5}$$

This establishes eq (2 22)

From (A.5) we have

$$\frac{1}{\pi i} \int_{l_0}^{\infty} dl\, a(l, s)\, \mathcal{F}(l, \beta)=\frac{1}{\beta} \hat{O}(\beta) \sum_{n=0}^{\infty} n\, a(\tfrac{1}{2} n-\tfrac{1}{2}, s) \frac{\sin \pi(\beta-n)}{\pi(\beta-n)}, \qquad (l_0>-1),$$

$$=\frac{1}{2\pi} \frac{1}{\beta} \hat{O}(\beta) \int_{-\pi}^{\pi} d\zeta\, e^{-i\beta\zeta} \sum_{n=0}^{\infty} n\, a(\tfrac{1}{2} n-\tfrac{1}{2}, s)\, e^{in\zeta}$$

$$=-\frac{i}{2\pi} \frac{1}{\beta} \hat{O}(\beta) \int_{-\pi}^{\pi} d\zeta\, e^{-i\beta\zeta} \frac{d}{d\zeta} f(\zeta), \tag{A 6}$$

where

$$f(\zeta) \equiv \sum_{n=0}^{\infty} a(\tfrac{1}{2} n-\tfrac{1}{2}, s)\, e^{in\zeta} \tag{A.7}$$

The right-hand side of (A 6)

$$= -\frac{i}{2\pi}\frac{1}{\beta}\hat{O}(\beta)[\mathrm{e}^{-i\beta\pi}f(\pi) - \mathrm{e}^{i\beta\pi}f(-\pi) + i\beta\int_{-\pi}^{\pi}\mathrm{d}\zeta\ \mathrm{e}^{-i\beta\zeta}f(\zeta)] \tag{A.8}$$

Since

$$\begin{aligned}\hat{O}(\beta)\,\mathrm{e}^{\pm i\beta\pi} &= \mathrm{e}^{\pm\beta\sinh i\pi}\\ &= 1 \qquad ,\end{aligned}$$

and $f(-\pi) = f(\pi)$, the first two terms in the square bracket cancel each other We therefore obtain

$$\begin{aligned}\frac{1}{\pi i}\int_{l_0}^{\infty}\mathrm{d}l\, a(l,s)\,\mathcal{F}(l,\beta) &= \frac{1}{2\pi}\frac{1}{\beta}\hat{O}(\beta)\,\beta\int_{-\pi}^{\pi}\mathrm{d}\zeta\ \mathrm{e}^{-i\beta\zeta}f(\zeta)\\ &= \frac{1}{\beta}\hat{O}(\beta)\,\beta\sum_{n=0}^{\infty}a(\tfrac{1}{2}n-\tfrac{1}{2},s)\frac{\sin\pi(n-\beta)}{\pi(n-\beta)}\end{aligned} \tag{A 9}$$

Eq (A 5) also gives

$$\begin{aligned}\mathcal{F}(-l-1,\beta) &= 2i\frac{1}{\beta}\sum_{n=-\infty}^{\infty}n\,\delta(2l+1+n)\,\hat{O}(\beta)\frac{\sin\pi(\beta-n)}{\beta-n}\\ &= -\frac{2i}{\beta}\sum_{n=-\infty}^{\infty}n\,\delta(2l+1-n)\,\hat{O}(\beta)\frac{\sin\pi(\beta+n)}{\beta+n}\ .\end{aligned} \tag{A 10}$$

Using (A.10) we then get

$$\begin{aligned}\frac{1}{\pi i}\int_{l_0}^{\infty}\mathrm{d}l\, a(l,s)\,\mathcal{F}(-l-1,\beta) &= -\frac{1}{\beta}\hat{O}(\beta)\sum_{n=0}^{\infty}n\,a(\tfrac{1}{2}n-\tfrac{1}{2},s)\frac{\sin\pi(\beta+n)}{\pi(\beta+n)}\\ &= \frac{i}{2\pi}\frac{1}{\beta}\hat{O}(\beta)\int_{-\pi}^{\pi}\mathrm{d}\zeta\ \mathrm{e}^{i\beta\zeta}\frac{\mathrm{d}f(\zeta)}{\mathrm{d}\zeta}\\ &= \frac{1}{\beta}\hat{O}(\beta)\,\beta\sum_{n=0}^{\infty}a(\tfrac{1}{2}n-\tfrac{1}{2},s)\frac{\sin\pi(n+\beta)}{\pi(n+\beta)}\end{aligned} \tag{A.11}$$

Adding (A 9) and (A 11) we obtain

$$h_1(s,\beta) = \frac{1}{\beta}\hat{O}(\beta)\,\beta\sum_{n=0}^{\infty}a(\tfrac{1}{2}n-\tfrac{1}{2},s)\left[\frac{\sin\pi(n-\beta)}{\pi(n-\beta)} + \frac{\sin\pi(n+\beta)}{\pi(n+\beta)}\right], \tag{A.12}$$

which is eq (2 23) in the text

Eq (A.12) has to be compared with Wallace's FB amplitude $\widetilde{h}_1(s,\beta)$

$$\widetilde{h}_1(s,\beta) = \frac{1}{\beta}\hat{O}(\beta)\,\beta\, a(\tfrac{1}{2}\beta-\tfrac{1}{2}, s) \tag{A.13}$$

Let us consider the difference

$$\sum_{n=0}^{\infty} a(\tfrac{1}{2}n-\tfrac{1}{2}, s)\left[\frac{\sin\pi(n-\beta)}{\pi(n-\beta)} + \frac{\sin\pi(n+\beta)}{\pi(n+\beta)}\right] - a(\tfrac{1}{2}\beta-\tfrac{1}{2}, s)$$

as a function of β Then this function vanishes for positive integral values of β and is regular in some positive β half-plane. However, it does not satisfy the boundedness condition of Carlson's theorem [18] Hence it cannot vanish identically. Therefore, $h_1(s,\beta) \neq \widetilde{h}_1(s,\beta)$ In other words, the interchange of $\hat{O}$ and the integration in (2 13) is not allowed.

We now show that $h_1(s,\beta) \to \widetilde{h}_1(s,\beta)$ when $\beta \to \infty$ To this end let us determine the asymptotic form of the infinite sum on the right-hand side of (A 12)

$$\begin{aligned}
&\sum_{n=0}^{\infty} a(\tfrac{1}{2}n-\tfrac{1}{2}, s)\left[\frac{\sin\pi(n-\beta)}{\pi(n-\beta)} + \frac{\sin\pi(n+\beta)}{\pi(n+\beta)}\right] \\
&= \sum_{\nu} a(p\nu-\tfrac{1}{2}, s)\left[\frac{\sin 2p\pi(\nu-b)}{2p\pi(\nu-b)} + \frac{\sin 2p\pi(\nu+b)}{2p\pi(\nu+b)}\right], \quad (\beta = 2pb,\ n = 2p\nu), \\
&= \sum_{\nu} \Delta\nu\, a(p\nu-\tfrac{1}{2}, s)\left[\frac{\sin 2p\pi(\nu-b)}{\pi(\nu-b)} + \frac{\sin 2p\pi(\nu+b)}{\pi(\nu+b)}\right], \quad \left(\Delta\nu = \frac{1}{2p}\right), \\
&\xrightarrow[p\to\infty]{} \int_0^{\infty} \mathrm{d}\nu\, a(p\nu-\tfrac{1}{2}, s)\,[\delta(\nu-b) + \delta(\nu+b)] \\
&= a(pb-\tfrac{1}{2}, s) \\
&= a(\tfrac{1}{2}\beta-\tfrac{1}{2}, s)
\end{aligned} \tag{A.14}$$

From (A.12) and (A 13) it then follows that $h_1(s,\beta)$ approaches $\widetilde{h}_1(s,\beta)$ when β is asymptotic.

Appendix B

To establish eq. (3 7), we have to evaluate the FB integral,

$$I(y) \equiv \int_0^{\infty} \beta\mathrm{d}\beta\, J_0(\beta y)\, \Delta(l,\beta^2),$$

of the distribution $\Delta(l, \beta^2)$ defined in eq (3.6a) First we consider $y > 1$

$$I(y) = \int_0^\infty \beta \mathrm{d}\beta\, J_0(\beta y)\, \beta^{2l+1} J_{-2l-1}(\beta) \frac{\Gamma(-l)}{2\pi i} \int_\infty^{0+} \frac{\mathrm{d}\lambda}{\lambda^2} (-\lambda)^{-l} \mathrm{e}^{-\beta^2/(\lambda+\delta)}$$

$$\text{(limit } \delta \to 0)$$

$$= \frac{2^{2l+1}}{y^2} \int_0^\infty x \mathrm{d}x\, J_0(x) \sum_{n=0}^\infty \frac{(-1)^n}{n!\,\Gamma(-2l+n)} \left(\frac{x}{2y}\right)^{2n}$$

$$\times \frac{\Gamma(-l)}{2\pi i} \int_\infty^{0+} \frac{\mathrm{d}\lambda}{\lambda^2} (-\lambda)^{-l} \mathrm{e}^{-x^2/(\lambda y^2 + a)} \qquad (a = y^2 \delta \to 0)$$

$$= \frac{2^{2l}}{y^2} \sum_{n=0}^\infty \frac{(-1)^n}{\Gamma(-2l+n)} \frac{1}{(4y^2)^n} \frac{\Gamma(-l)}{2\pi i} \int_\infty^{0+} \frac{\mathrm{d}\lambda}{\lambda^2} (-\lambda)^{-l}$$

$$\times (\lambda y^2 + a)^{n+1} \mathrm{e}^{-(\lambda y^2 + a)/4}\, {}_1F_1(-n, 1, \tfrac{1}{4}(\lambda y^2 + a)) \qquad \text{(ref [23])}$$

$$= -\sum_{n=0}^\infty \frac{1}{\Gamma(-2l+n)} \frac{\Gamma(-l)}{2\pi i} \int_\infty^{0+} \mathrm{d}\lambda (-\lambda)^{n-l-1} \mathrm{e}^{-\lambda y^2}\, {}_1F_1(-n, 1, \lambda y^2)$$

$$\text{(taking limit } a \to 0)$$

$$= -\sum_{n=0}^\infty \frac{1}{\Gamma(-2l+n)} \sum_{m=0}^n \frac{n!\, y^{2m}}{(n-m)!\,(m!)^2} \frac{\Gamma(-l)}{2\pi i} \int_\infty^{0+} \mathrm{d}\lambda (-\lambda)^{n+m-l-1} \mathrm{e}^{-\lambda y^2}$$

$$= \frac{y^{2l}}{\Gamma(l+1)} \sum_{n=0}^\infty \frac{(-1)^n\, n!}{\Gamma(-2l+n)} \frac{1}{y^{2n}} \sum_{m=0}^n \frac{(-1)^m\, \Gamma(n+m-l)}{(n-m)!\,(m!)^2} \qquad \text{(ref. [24])}$$

$$= \frac{y^{2l}}{\Gamma(l+1)} \sum_{n=0}^\infty \frac{(-1)^n\, \Gamma(n-l)}{\Gamma(-2l+n)} \frac{1}{y^{2n}}\, {}_2F_1(-n, n-l, 1, 1)$$

$$= \frac{y^{2l}}{\Gamma(l+1)} \sum_{n=0}^\infty \frac{(-1)^n\, \Gamma(n-l)}{\Gamma(-2l+n)} \frac{1}{y^{2n}} \frac{\Gamma(l+1)}{n!\,\Gamma(l+1-n)} \qquad \text{(ref [25])}$$

$$= y^{2l} \frac{\Gamma(-l)}{\Gamma(-2l)\,\Gamma(l+1)}\, {}_2F_1(-l, -l, -2l, 1/y^2) \qquad (y > 1)$$

$$= -\frac{2}{\pi} \sin \pi l\, Q_{-l-1}(2y^2 - 1) \tag{B 1}$$

For $y < 1$,

$$I(y) = \int_0^\infty \mathrm{d}\beta \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} (\tfrac{1}{4}y^2)^n \beta^{2n+2l+2} J_{-2l-1}(\beta)$$

$$\times \frac{\Gamma(-l)}{2\pi i} \int_\infty^{0+} \frac{\mathrm{d}\lambda}{\lambda^2} (-\lambda)^{-l} \mathrm{e}^{-\beta^2/(\lambda+\delta)}$$

$$= 4^l \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} (\tfrac{1}{4}y^2)^n \frac{1}{\Gamma(-2l)} \frac{\Gamma(-l)}{2\pi i} \int_\infty^{0+} \frac{\mathrm{d}\lambda}{\lambda^2} (-\lambda)^{-l} (\lambda+\delta)^{n+1}$$

$$\times \mathrm{e}^{-(\lambda+\delta)/4} {}_1F_1(-2l-n-1, -2l, \tfrac{1}{4}(\lambda+\delta)) \qquad (\text{ref [23]})$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(-2l)} y^{2n} \sum_{m=0}^{\infty} \frac{\Gamma(-2l-n-1+m)\Gamma(-2l)}{\Gamma(-2l-n-1)\,\Gamma(-2l+m)} \frac{1}{m!}$$

$$\times \frac{\Gamma(-l)}{2\pi i} (-1)^{n+m+1} \int_\infty^{0+} \mathrm{d}\lambda (-\lambda)^{m+n-l-1} \mathrm{e}^{-\lambda}$$

(taking $\delta \to 0$ limit and expanding the hypergeometric function)

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(-2l)\,\Gamma(l+1)} y^{2n} \sum_{m=0}^{\infty} \frac{\Gamma(-2l-n-1+m)}{\Gamma(-2l-n-1)} \frac{\Gamma(-2l)}{\Gamma(-2l+m)} \frac{1}{m!}$$

$$\times \Gamma(m+n-l) \qquad (\text{Ref [24]})$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(-2l)\,\Gamma(l+1)} y^{2n} \Gamma(n-l)\, {}_2F_1(-2l-n-1, n-l, -2l, 1)$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \frac{\Gamma(n-l)}{\Gamma(n+1)\,\Gamma(-l-n)} y^{2n} \qquad (\text{ref [25]})$$

$$= {}_2F_1(l+1, -l, 1, y^2) \qquad (y < 1)$$

$$= P_l(1-2y^2) \qquad \text{(B 2)}$$

We have explicitly shown above that

$$\int_0^\infty \beta \mathrm{d}\beta\, J_0(\beta y)\, \Delta(l, \beta^2) = -\frac{2}{\pi} \sin \pi l\, Q_{-l-1}(2y^2-1) \qquad (y > 1) \qquad \text{(B 1)}$$

$$= P_l(1-2y^2) \qquad (y < 1) \qquad \text{(B 2)}$$

Furthermore, we have the following result [22]

$$\int_0^\infty \beta \mathrm{d}\beta\, J_0(\beta y) \frac{J_{2l+1}(\beta)}{\beta} = -\frac{2}{\pi} \sin \pi l\, Q_l(2y^2-1) \qquad (y>1) \tag{B.3}$$

$$= P_l(1-2y^2) \qquad (y<1) \tag{B 4}$$

From eqs (B 1–B.4) it follows that

$$\int_0^\infty \beta \mathrm{d}\beta\, J_0(\beta y) \left[\Delta(l, \beta^2) - \frac{J_{2l+1}(\beta)}{\beta}\right] \frac{1}{2\cos \pi l}$$

$$= \frac{\theta(y-1)}{2\cos \pi l} \left[\frac{2}{\pi} \sin \pi l\, Q_l(2y^2-1) - \frac{2}{\pi} \sin \pi l\, Q_{-l-1}(2y^2-1)\right]$$

$$= P_l(2y^2-1)\, \theta(y-1)\,, \tag{B.5}$$

which proves eq. (3 7).

We now show how eq. (3 6b) is obtained from eq (3 6a)

$$\frac{\Gamma(-l)}{2\pi i} \int_\infty^{0+} \frac{\mathrm{d}\lambda}{\lambda^2} (-\lambda)^{-l}\, \mathrm{e}^{-\beta^2/(\lambda+\delta)}$$

$$= \frac{\Gamma(-l)}{2\pi i} \int_\infty^{0+} \frac{\mathrm{d}\lambda}{\lambda^2} (-\lambda)^{-l} \tfrac{1}{2}(\lambda+\delta) \int_0^\infty x \mathrm{d}x\, J_0(x\beta)\, \mathrm{e}^{-(\lambda+\delta)x^2/4}$$

$$= -2 \int_0^\infty x\mathrm{d}x\, J_0(x\beta) \frac{\mathrm{d}}{\mathrm{d}x^2} \left[\mathrm{e}^{-\delta x^2/4} \frac{\Gamma(-l)}{2\pi i} \int_\infty^{0+} \frac{\mathrm{d}\lambda}{\lambda^2} (-\lambda)^{-l}\, \mathrm{e}^{-\lambda x^2/4}\right]$$

$$= -2 \frac{\Gamma(-l-1)}{\Gamma(l+1)} \int_0^\infty x\mathrm{d}x\, J_0(x\beta) \frac{\mathrm{d}}{\mathrm{d}x^2} [\mathrm{e}^{-\delta x^2/4} (\tfrac{1}{4}x^2)^{l+1}]$$

$$= \frac{\Gamma(-l)}{\Gamma(l+2)} \frac{1}{\beta^{2(l+1)}} \int_0^\infty \mathrm{d}x\, J_1(x)\, \mathrm{e}^{-\delta x^2/4\beta^2} (\tfrac{1}{4}x^2)^{l+1} \tag{B.6}$$

Using (B.6) in (3.6a) gives (3 6b) It is instructive to point out that the integration on the right-hand side of (B 6) can be carried out further, leading to the result

$$\frac{\Gamma(-l)}{2\pi i} \int_\infty^{0+} \frac{\mathrm{d}\lambda}{\lambda^2} (-\lambda)^{-l}\, \mathrm{e}^{-\beta^2/(\lambda+\delta)}$$

$$= \frac{1}{\beta^{2(l+1)}} \left[\Gamma(-l) \left(\frac{\beta^2}{\delta}\right)^{l+2} e^{-\beta^2/\delta} {}_1F_1\left(-l, 2, \frac{\beta^2}{\delta}\right)\right] \tag{B.7}$$

As expected, in the limit $\delta \to 0$ the right-hand side of (B 7) becomes $\beta^{-2(l+1)}$

References

[1] F Halzen, Particle interactions at very high energies, ed D Speiser, F Halzen and J Weyers, vol 4A (Plenum, New York, 1974) p 1
[2] G Cohen-Tannoudji, Proc 10the session Herceg-Novi Summer School, Basko-Polje, 1974
[3] W Czyz, Advances in nuclear physics, ed M Baranger and E Vogt, vol 4 (Plenum, New York, 1971) p 61,
D R Yennie, Hadronic interactions of electrons and photons, ed J Cumming and H Osborn (Academic, London, 1971) p 321
[4] E Gerjuoy and B K Thomas, Reports on Progr in Phys 37 (1974) 1345
[5] R L Sugar, Lectures in theoretical physics, ed A O Barut and W E Brittin, vol 14A (Colorado University Press, 1972)
[6] H M Fried, Functional methods and models in quantum field theory (MIT Press, 1972)
[7] R J Glauber, Lectures in theoretical physics, ed W E Brittin and L G Dunham, vol 1 (Interscience, New York, 1959) p 315
[8] M Lévy and J Sucher, Phys Rev 186 (1969) 1656
[9] W N Cottingham and R F Peierls, Phys Rev. 137 (1965) B147
[10] T Adachi and T Kotani, Progr Theor Phys. Suppl (1965) 316, Progr Theor, Phys 35 (1966) 463, Progr Theor Phys Suppl (1966) 297
[11] E Predazzi, Ann of Phys 36 (1966) 228
[12] M M Islam, Lectures in theoretical physics, ed A O Barut and W E Brittin, vol 10B (Gordon and Breach, 1968) p 97
[13] S J Wallace, Phys Rev D8 (1973) 1846
[14] S J Wallace, Phys Rev D9 (1974) 406
[15] M L Goldberger and R Blankenbecler, Phys Rev 126 (1962) 766
[16] R G Newton, The complex j-plane (Benjamin, 1964)
[17] V de Alfaro and T Regge, Potential scattering (North-Holland, Amsterdam, 1965)
[18] P D B Collins and E J Squires, Regge poles in particle physics, vol 45 (Springer tracts in modern physics, 1968)
[19] B Schrempp and F Schrempp, Nucl Physics B60 (1973) 110
[20] Y L Luke, The special functions and their approximations, vol 1 (Academic, New York, 1969) p 34
[21] A Erdelyi, ed , Higher transcendental functions (Bateman manuscript project), vol 1 (McGraw-Hill, New York, 1954) p 155
[22] M Abramowitz and I A Stegun, ed , Handbook of mathematical functions (National Bureau of Standards, Washington, 1972) p 487
[23] A Erdélyi, ed , Tables of integral transforms (Bateman manuscript project), vol 2 (McGraw-Hill, New York, 1954) p 30
[24] E T Whittaker and G N Watson, Modern analysis (Cambridge University Press, 1965) p 245
[25] Higher transcendental functions, ref [21] vol 1, p 104
[26] J Kupsch and I O Stamatescu, Nuovo Cimento 15A (1973) 663
[27] J Kupsch and I O Stamatescu, Nuovo Cimento 20A (1974) 317